The empirical recurrence for OEIS A231905, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A231905 records “Empirical recurrence of order 81” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 153$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 83$. Its last coefficient is $c_{81} = -1100864 \neq 0$, so no further tail satisfies anything shorter; any order-81 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231905 is “Number of nX4 0..2 arrays with no element having a strict majority of its horizontal, diagonal and antidiagonal neighbors equal to itself plus one mod 3, with upper left element zero (rock paper and scissors drawn positions).”. It has offset 1 and begins

$$8, 948, 31167, 1082472, 37368831, 1291573433, 44640322903, 1542901809201, \ \ldots$$

The entry states:

> Empirical recurrence of order 81 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:49:09, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 153$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 153$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 81: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 81.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 306$ exact terms from $n = 2$ on, it returns order 81 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{81} c_i \, a(n-i),$$

c_1	51	c_{22}	21025054845273	c_{43}	80029768323935213	c_{64}	2540969638770151
c_2	−595	c_{23}	35310308088591	c_{44}	168315751566503242	c_{65}	−1716563081891817
c_3	370	c_{24}	−69617395136375	c_{45}	23333356253170259	c_{66}	327568728243684
c_4	18356	c_{25}	−90336512084748	c_{46}	−132189430328425009	c_{67}	391494328996782
c_5	62648	c_{26}	214519484524829	c_{47}	−280867003976119269	c_{68}	−348421172677836
c_6	−965422	c_{27}	−110827924978975	c_{48}	251184577384081783	c_{69}	19875075576423
c_7	−1713432	c_{28}	−328471111506473	c_{49}	49277961316507194	c_{70}	47654404986313
c_8	21420548	c_{29}	769572659630006	c_{50}	115923043735869938	c_{71}	−21682956513182
c_9	53078217	c_{30}	307854505400884	c_{51}	−137324050479356428	c_{72}	8936284949820
c_{10}	−337031583	c_{31}	−809704135062611	c_{52}	−77417661454261349	c_{73}	3161617189663
c_{11}	−797850364	c_{32}	−2857448584556325	c_{53}	−21046505270583419	c_{74}	−558640272747
c_{12}	2663605845	c_{33}	7279696262411442	c_{54}	160658342738940006	c_{75}	192480150389
c_{13}	9019565974	c_{34}	−10275083690891566	c_{55}	−47401832115006446	c_{76}	21237608338
c_{14}	−8211231635	c_{35}	8319655730599587	c_{56}	−78109869755371690	c_{77}	−24583517497
c_{15}	−58781463604	c_{36}	9616101789535192	c_{57}	55075308117519960	c_{78}	2836411588
c_{16}	−105615098799	c_{37}	−16476787691997842	c_{58}	4667873446859935	c_{79}	203670492
c_{17}	293965571913	c_{38}	−63197366937990929	c_{59}	−20782312716811577	c_{80}	31808800
c_{18}	1085088177394	c_{39}	156826295207045377	c_{60}	15153692651931099	c_{81}	−1100864
c_{19}	422296214011	c_{40}	−49556831153692856	c_{61}	1832715448789769		
c_{20}	−8965857222899	c_{41}	−65091336724844443	c_{62}	−10262212028306065		
c_{21}	−1279035934237	c_{42}	−98509509065239707	c_{63}	2745018503365796		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 83$, and it is the recurrence OEIS A231905 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{81} - c_1 x^{81-1} - \cdots - c_{81}$. Its constant term is $-c_{81} = 1100864$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 81 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 81, so $\deg q = 81$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 81, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 81, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1100864.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-81 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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